Vijaya Krishna Reddy Polu

• 716-908-6112 • vijaykrishnareddy09@gmail.com• LinkedIn • GitHub

EDUCATION

University at Buffalo, The State University of New York – Buffalo, NY Aug 2024- Dec 2025

Master of Science in Engineering Science Data Science

- **Relevant Coursework:** Distributed Systems, Algorithm Design, Optimization Mathematics, Programming and Data Structures, Statistical Learning, Data Models and Query Languages, Machine Learning, Data-Intensive Computing.

TECHNICAL SKILLS

- **Programming Languages & Databases:** Java, Python, C, Data Structures, R, MySQL, ETL Pipelines.
- **Web Technologies:** HTML, CSS, JS.
- **Frameworks:** Pyspark, Spring MVC, Hibernate, Scikit-learn, NumPy, Matplotlib, TensorFlow, Natural Language Toolkit.
- **Core Concepts:** Object-Oriented Design, Algorithm Design, Distributed Systems
- **Cloud & Data Platforms: AWS (RDS, Lambda, S3, Glue, Crawler),** Spark, Hadoop, Kafka.
- **Data Analytics & Visualization:** ETL, Power BI, Tableau, Matplotlib, Seaborn, Plotly.

WORK EXPERIENCE

❖ **Accenture Pvt Ltd, Bangalore, India – Senior Software Engineer** May 2023 – Aug 2024

Client: Kaiser Permanente

- Automated large-scale, high-performance healthcare workflows using **Java, Python, and IBM TWS**, reducing manual intervention and operational errors by 75%, while improving workflow reliability across distributed systems.
- Designed and implemented **fault-tolerant job dependency models and optimized scheduling algorithms**, enabling real-time low-latency responsiveness, reducing downtime by 30%, and boosting overall system throughput.
- Built a centralized monitoring and anomaly detection toolkit leveraging Java, SQL and shell scripting, which reduced incident resolution time by 60 % and enabled **proactive, intelligent, scalable** root cause analysis.
- Developed **scalable, reusable automation frameworks for hundreds of scheduling scenarios**, accelerating deployments, improving maintainability, and ensuring high availability of healthcare operations.

❖ **Accenture Pvt Ltd, Bangalore, India – Software Engineer** Sep 2021 – May 2023

- Integrated healthcare platform operations with real-time job orchestration tools, improving **system-wide, end-to-end, enterprise-grade** visibility and operational control for IT teams and reducing production issues significantly.
- Participated in Agile sprints to design and implement backend features using Java and SQL, aligning **scalable, business-critical** system architecture with business priorities and accelerating deployment cycles across environments.
- Built proactive monitoring dashboards and automated log analysis solutions in Python and SQL, detecting anomalies early and enabling **predictive, high availability** improving system uptime by 25% while boosting team productivity.

PROJECTS

❖ **NYC Yellow Taxi Demand Forecasting Pipeline – *Technologies: AWS (Lambda,Glue, S3, Athena, RDS), LightGBM, Streamlit***

- Architected and deployed a large-scale, distributed AWS pipeline leveraging Lambda, Glue, and S3 to process millions of taxi trip records daily, engineered optimized ETL flows with feature selection and lag variables, and reduced data processing latency by 40% while ensuring system scalability and fault tolerance.
- Designed and implemented an interactive Streamlit dashboard hosted on AWS Elastic Beanstalk, integrating Athena and RDS for real-time querying, forecasting, and visualization, which accelerated operational decision-making by 60% and provided stakeholders with actionable insights into city-wide demand patterns.

❖ **Weather Forecasting Using ML Models – *Technologies: Python, Scikit-learn, SVM, XGBoost***

- Developed end-to-end **Machine learning pipelines** in Python using Scikit-learn, SVM, and XGBoost for real-time weather forecasting, achieving an **F1-score of 0.88.** Designed scalable, modular preprocessing, feature engineering, and model selection workflows with **hyperparameter tuning and cross-validation** for deployment-ready applications.

❖ **Hybrid Models for Aviation Risk Prediction – *Technologies: Python, SVM, DNN, TF-IDF***

- Developed a **Hybrid SVM and DNN model** in **Python** for **aviation risk prediction**, achieving **92% accuracy** on structured and unstructured flight data. Engineered **TF-IDF features**, implemented **modular preprocessing pipelines**, and validated model robustness via **probabilistic decision trees, tokenization, and cross-validation**, ensuring **reliable, scalable, deployment-ready predictions**.

❖ **Analysis of Chest X-rays for Automated Disease Detection - Technologies: Python, PyTorch, TensorFlow, ResNet50, EfficientNet-B0, Vision Transformer (ViT), MLP, Streamlit, Grad-CAM, t-SNE, PCA**

- Engineered a **hybrid multi-label classification system** in **Python** using **ResNet50, EfficientNet-B0, and ViT** to detect 15 thoracic diseases, integrating **image features with patient metadata**. Applied **focal loss, class-balanced sampling, and perturbations** for robust predictions on 112k X-rays. Optimized architectures with **LeakyReLU, dropout, and hyperparameter tuning**, achieving **macro F1 0.65 and AUROC 0.84**, with **Grad-CAM, t-SNE, PCA** interpretability. Deployed a **scalable Streamlit web application** for **real-time inference** (87ms CPU), demonstrating **production-ready, end-to-end pipeline engineering**.